

~~2020~~

①

(1)

What is the effect of dilution on Specific Conductance?

Give Reason: Specific conductance of the solution is the conductance due to one centimeter cube of the solution. Specific conduction and dilution is inversely connected.

Reason: As the specific conductance increases dilution decreases and vice versa as the presence of a number of ions in 1ml decreases.

Due to dilution, the volume of solution increases and there is increase in the number of ions. But the number of ions per cc of solution decreases. This is the reason specific conductivity of solution decreases on dilution.

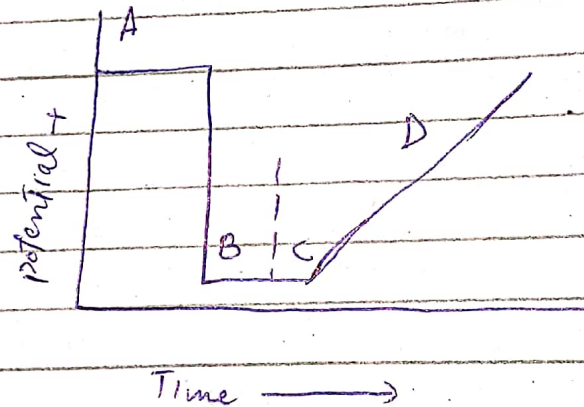
Note: The equivalent conductivity and molar conductivity increases with dilution. The equivalent and molar conductivities tend to accumulate maximum value with increasing dilution.



(ii)

What is the basic Principle of Anodic stripping Voltammetry

Anodic stripping voltammetry is a voltammetric method for quantitative determination of specific ionic species. The analyte of interest is electroplated on the working electrode during a deposition step and oxidized from the electrode during the stripping step. The current is measured during the stripping step. The oxidation of species is registered as a peak in the current signal at the potential at which the species begin to oxidize. The stripping step can either be linear, staircase, squarewave or pulse.



(ii)

What is Ilkovic equation? Give its significance

It is ~~the~~ a relation used in polarography relating the diffusion current (I_d) and the concentration of the depolarizer (C) which is the substance reduced or oxidized at the dropping mercury electrode.

OR It is the linear relationship b/w the diffusion current and the concentration of electroactive species

$$I_d = 607 n C D^{1/2} m^{2/3} t^{1/6}$$

I_d = Diffusion current in μA , n = no. of electrons

D = Diffusion Coefficient C = concentration m = mass of mercury

Significance: The application of the Ilkovic equation to quantitative analysis was shown to be inadvisable unless it is definitely established that experimental conditions are well within the range for insuring diffusion control.

A logarithmic relation between critical drop time and suppressor concentration was found.

(iv)

Give application of Conductometry in Precipitation titration?

If the electrical conductivity of the solution is recorded during the titration we refer to this as conductometric titration. For precipitation conductometric titration whether or not the conductance of a solution decreases as the precipitate forms will depend on the relative mobility of the ions in solution.

Precipitation conductometric titration can be used to determine the concentration of an electrolytic solution e.g. Precipitation titration of KCl with AgNO_3 . It is used for determination of halide ions in the solution. It is used to measure salt content in food beverages and water.

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(v)

Specific / molar Conductance (Repeat)

(iv)

What do you mean by Cyclic Voltammetry?

Cyclic voltammetry is an electrochemical technique for measuring the current response of a redox active solution to a linearly cycled potential sweep between two or more set values.

It is useful method for quickly determining information about the thermodynamics of redox processes, the energy levels of the analyte and the kinetics of electronic transfer reactions.

(vii)

Changes in Conductance in the titration mixture of HCl and acetic acid with NaOH.

HCl and NaOH as a strong acid and base. ~~when~~ Before ~~NaOH~~ is added the conductance is high due to the presence of highly mobile H^+ ion. When the base is added the conductance falls due to the replacement of hydrogen ion by the added cations as

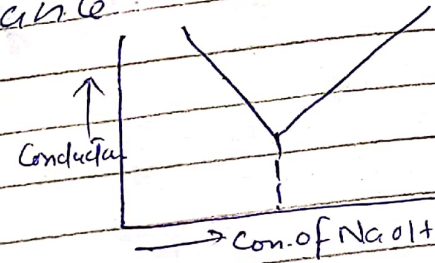
increasing dilution ~ accumulate ~ equivalent

ions react with OH^- ions to form undissociated water. This decrease in the conductance continues till the equivalence point.

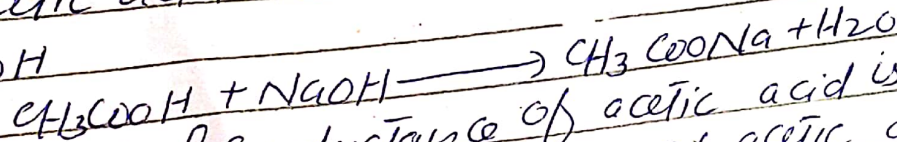
At the equivalence point the solution contains only NaCl . After the equivalence point the conductance increases due to large conductivity of OH^- ions.

CH_3COOH vs NaOH

Acetic acid neutralised with strong base



NaOH



- * The initial conductance of acetic acid is low.
- * As NaOH is added H^+ ions of acetic acid are replaced by Na^+ ions from NaOH .
- * This results in the formation of sodium acetate (CH_3COONa) which has higher conductance than acetic acid.
- * Due to this slight increase in conductance is observed in the middle portion.
- * After the equivalence point conductance highly increases due to high conductivity of OH^- ions.

OR

OR When NaOH is added to a solution of HCl the concentration of hydroxyl ions (OH^-) will increase after the endpoint is reached thereby increasing the conductivity.

Initially the conductance is low due to the feeble ionization of acetic acid followed by increase due to NaOH

Beyond the equivalence point steep rises due to excess of NaOH

(VIII)

What is Cell Constant? What is relation between Conductivity of Cell and Cell Constant

The cell constant is a constant value for particular cell that represents the ratio of distance between electrodes and the cross section area of electrode in a conductance cell

$$\text{Cell Constant} = \frac{\text{Distance between electrodes}}{\text{Cross section area of electrodes}}$$

We know that

$$\text{Specific Conductance } K = \frac{1}{R} \cdot \frac{1}{A}$$

$$K = \text{observed Conductance} \times \text{Cell Constant}$$

$$\text{Cell Constant} = \frac{\text{Specific Conductance}}{\text{observed Conductance}}$$

Summary

(ix)

Arc

or

Arc is an electrical breakdown of a gas that produces a prolonged electrical discharge.

It comes from one conductor

It remains for long time

It works on the principle of continuous electrical discharge.

Spark

An electric spark is an abrupt electrical discharge when electric field strength exceeds $\text{app} = 30 \text{ kV/cm}$

spark comes from two conductors

It remains for small time

It works on the principle of current pulse method.

(3) (2020)

(X)

Give three advantages of DME?

- * Mercury form amalgam with most metals
- * Mercury has a high hydrogen overvoltage
- * It provides a smooth fresh surface for the reaction
- * Each drop is remain unaffected and does not become contaminated by the deposited metal
- * Diffusion equilibrium is readily established at mercury solution interface

